Supplemental Material for "Two-channel signatures of the E_7 spectrum in a blockaded Rydberg chain"

OpenQuantum

This supplement collects the model, mathematical relations, numerical definitions, and completed checks underlying the main text. Figures 1–3 refer to the main article; sections, equations, and tables here carry the prefix S. References are numbered independently.

S1. MODEL, FIELD CONTENT, AND CRITICAL CALIBRATION

We study the strict-blockade Hamiltonian

$$H = \sum_j \left[\frac{\Omega}{2} P_{j-1} X_j P_{j+1} - \Delta n_j\right] + \sum_{i<j,\, d_{ij}\geq 2} V_2 (2/d_{ij})^6 n_i n_j, \qquad n_j n_{j+1} = 0, \qquad P_j = 1 - n_j, \tag{S1}$$

with $V_2 < 0$. This constrained space connects hard-boson models [1] and blockaded-spin descriptions [2]. For open chains, $d_{ij} = |i-j|$ and missing endpoint projectors are the identity; for periodic chains, d_{ij} is the shortest ring distance. Every allowed pair is counted once. We use $\hbar = \Omega = a = 1$, with a the lattice spacing.

Equation (S1) approximates the sign-reversed experimental problem in Ref. [3]. It omits the finite-blockade correction H_2, including diagonal shifts and correlated hopping of order Ω^2/V_1. No additional boundary detuning is included in the open-chain calculations. Within the projected model, Hamiltonian sign reversal maps the upper spectral edge to a ground-state problem; the sign of the transverse drive can be changed by a local phase rotation without changing the density observable.

The leading thermal perturbation of the $c = 7/10$ tricritical Ising (TCI) conformal field theory produces the integrable E_7 spectrum [4–6]. Writing $r_a = m_a/m_1$,

$$(r_1, \ldots, r_7) = (1,\ 1.2856,\ 1.8794,\ 1.9696,\ 2.5321,\ 2.8794,\ 3.7017), \tag{S2}$$

with internal $\mathbb{Z}_2$ parities $(-,+,-,+,+,-,+)$ in the disordered phase [7]. One-site translation exchanges the two density-wave patterns, implementing the internal symmetry together with spatial translation [8]. The long-wavelength density expansion is

$$n_j \simeq \bar{n} + c_\varepsilon \varepsilon + c_{\varepsilon'} \varepsilon' + (-1)^j (c_\sigma \sigma + c_{\sigma'} \sigma') + \cdots . \tag{S3}$$

Consequently, even zero-momentum continuum particles occur near lattice momentum zero, and odd ones near π. This uses bulk translation symmetry; open-chain spatial reflection is a different quantum number.

The total scaling dimensions are $x_\sigma = 3/40$, $x_{\sigma'} = 7/8$, $x_\varepsilon = 1/5$, and $x_{\varepsilon'} = 6/5$. Both detuning and interaction changes generate the even relevant perturbations $g_\varepsilon \varepsilon + g_{\varepsilon'} \varepsilon'$. Their renormalization-group eigenvalues are $y_\varepsilon = 9/5$ and $y_{\varepsilon'} = 4/5$. For a generic straight path of distance t from TCI,

$$\eta = \frac{g_{\varepsilon'}}{|g_\varepsilon|^{4/9}} \propto t^{5/9}, \qquad m_1 \propto t^{5/9}, \tag{S4}$$

provided the linear component of g_ε is nonzero. The first power follows from $y_{\varepsilon'}/y_\varepsilon = 4/9$ and $g_\varepsilon, g_{\varepsilon'} \propto t$; the mass power is $1/y_\varepsilon = 5/9$. Approaching the scaling limit also requires increasing L

with the correlation length. The map from lattice parameters to these two couplings has not been calibrated, so small mass-ratio errors do not establish a pure leading-thermal trajectory.

Following finite-size phenomenological scaling [9], we locate a TCI estimate using simultaneous crossings of L times the lowest odd and excited even gaps. The $L = 28/32$ crossing gives $(\Delta_c, V_{2c}) = (-1.5011, -1.6260)$, within 10^{-3} of the 24/28 estimate and close to the experimental calibration [3]. At criticality, the conformal gap relation [10] reads

$$\mathcal{E}_n - \mathcal{E}_0 \simeq \frac{2\pi v}{L} x_n. \tag{S5}$$

Calibrating with $x_\sigma = 3/40$ gives $v \simeq 1.90$. Other rescaled gaps are 0.875, 0.198, and 1.18, near 7/8, 1/5, and 6/5. This critical calibration is not an independent velocity measurement in the gapped regime.

S2. RESPONSE, WEIGHTS, AND NUMERICAL CONVENTIONS

For a periodic ground state, the connected density spectrum entering linear response [11] is

$$S_k(\omega) = \frac{1}{L} \sum_{n \neq 0} \left| \left\langle n \left| \sum_j e^{ikj} n_j \right| 0 \right\rangle \right|^2 \delta[\omega - (\mathcal{E}_n - \mathcal{E}_0)]. \tag{S6}$$

The elastic ground-state term is excluded. Uniform and staggered drives select $k = 0$ and π exactly on periodic chains. Their leading continuum operators are ε and σ, respectively; the staggered coupling was proposed in Ref. [3]. For a pure continuum operator $\mathcal{O}$, one-particle weights obey

$$w_a \propto \frac{|F_a^{\mathcal{O}}|^2}{m_a}. \tag{S7}$$

Known form factors [6, 12, 13] give $w_3/w_1 = 0.067$ and $w_6/w_1 = 0.0030$ for σ, and $w_4/w_2 = 0.143$, $w_5/w_2 = 0.038$, and $w_7/w_2 = 0.0012$ for $\Theta \propto \varepsilon$. Absolute interchannel amplitudes are not universal. Disorder-field matrix elements cannot be substituted for the energy-density matrix elements of the uniform probe.

Sparse exact diagonalization is performed in the constrained space, for periodic $L \leq 36$ and open $L = 19$–28. Reorthogonalized Lanczos spectra [14, 15] use up to 400 iterations. The light levels agree with direct eigensolver results to 10^{-6}; small systems agree with full-space diagonalization to machine precision. The first two gaps at $k = \pi$ are labelled $\widehat{m}_1, \widehat{m}_3$, and the first two excited gaps at $k = 0$ are $\widehat{m}_2, \widehat{m}_4$. These labels define candidates by sector and energy order. They do not independently identify particles. We write

$$\widehat{r}_a = \frac{\widehat{m}_a}{\widehat{m}_1}, \qquad e_a = 100 \left(\frac{\widehat{r}_a}{r_a} - 1 \right), \tag{S8}$$

where e_a is the error relative to the ideal E_7 ratio, in percent. Open-chain dominant line energies are denoted ω_a^{o}.

An isolated stable particle has a nonzero thermodynamic spectral weight in Eq. (S6). The finite-volume normalization of multiparticle matrix elements [16] implies that an individual two-particle level typically has weight proportional to $1/L$ away from threshold. Threshold form factors can alter this behavior; the reference curve in main-text Fig. 3 is not a fitted scaling law.

S3. MATCHING REGION AND FINITE-SIZE DIAGNOSTICS

The scans use

$$(\Delta, V_2) = (\Delta_c, V_{2c}) + t(\cos\theta, \sin\theta), \qquad \delta_L = 100 \max_{a=2,3,4} \left| \frac{\widehat{r}_a}{r_a} - 1 \right|. \tag{S9}$$

Main-text Fig. 1(a) uses a 21×21 grid at $L = 32$, spanning $\Delta \in [-2.00, -1.20]$ and $V_2 \in [-1.75, -1.05]$, with spacings 0.04Ω and 0.035Ω. Blank cells fail $L\widehat{m}_1/\Omega \geq 18$ or $\widehat{m}_2 \geq \widehat{m}_1/2$. These numerical filters do not establish phases or convergence. The 3% contour is a finite-chain matching region, not an experimental tolerance guarantee.

Along $\theta = 110^\circ$ at $L = 36$, $t = 0.1$–0.2 gives maximal errors below 3%, while $t = 0.3$ gives 3.01%. Directions 90° and 130° contain matching points; 180° never reaches 5% within the scan. At $\theta = 110^\circ, t = 0.2$, the reference point is

$$(\Delta, V_2) = (-1.5695, -1.4381), \qquad \widehat{m}_1(36) = 0.9196\Omega, \qquad (\widehat{r}_2, \widehat{r}_3, \widehat{r}_4) = (1.2678, 1.9243, 1.9508). \tag{S10}$$

The relative errors are $(-1.39, +2.39, -0.95)\%$. At $t = 0.1$, $\widehat{r}_3$ changes from 1.792 to 1.887 between $L = 28$ and 36; at $t = 0.3$, all three ratios change by less than 0.24% over the same sizes. These results concern finite chains, not a controlled limit of Eq. (S4).

In the ideal theory, A_4 is a shallow A_1A_1 bound state. Its inverse binding length is set by [12, 17]

$$\mu_{11}^4 = \sqrt{m_1^2 - m_4^2/4} = m_1 \sin(\pi/18) \simeq 0.174 m_1. \tag{S11}$$

The leading bound-state finite-volume correction can scale as $\exp(-\mu_{11}^4 L/v)$, much more slowly than $\exp(-m_1 L/v)$ [17, 18]. With the working gap and critical velocity, $v/\mu_{11}^4 \simeq 12$ sites, whereas $v/m_1 \simeq 2$ sites. These are diagnostic estimates using a velocity not measured independently at the working point.

Define the dimensionless and dimensional bindings by

$$b_4 = 2 - \widehat{r}_4, \qquad B_4 = 2\widehat{m}_1 - \widehat{m}_4. \tag{S12}$$

At $t = 0.2$, b_4 decreases from 0.159 at $L = 18$ to 0.0492 at $L = 36$. The latter is about 62% above the ideal $2 - r_4 = 0.0304$. Existing fits to $b_4^\infty + c\exp(-L/\ell)$ have drifting lengths: ℓ increases from 4.5 to 7.2 sites as the lower fitting cutoff increases from $L = 22$ to 30. The corresponding fitted intercepts change from approximately 0.04765 to 0.04401. The windows establish neither an asymptotic intercept nor a unique nonintegrable contribution to the binding.

From $L = 24$ to 36, the candidate weights change by less than 0.4% for A_1, A_2, increase by 9.6% for A_3, and decrease by 11.2% for A_4. The next odd and even levels lose factors of 2.6 and 1.6, respectively. At $L = 36$, $w_3/w_1 = 0.0483$ and $w_4/w_2 = 0.129$, below their pure-field values. In Eq. (S3), the relative amplitude of σ' in the staggered probe scales as

$$\frac{c_{\sigma'}}{c_\sigma} \left(\frac{m_1 a}{v}\right)^{x_{\sigma'} - x_\sigma} = \frac{c_{\sigma'}}{c_\sigma} \left(\frac{m_1 a}{v}\right)^{4/5} \tag{S13}$$

at fixed CFT normalization. Operator mixing and finite-volume effects are not separated by the observed trends.

S4. FIXED-GAP DIFFERENTIAL RESPONSE

Form-factor perturbation theory [19] gives a more restrictive comparison than matching energy ratios at one point. In the normalization of Ref. [7], the response to the subleading thermal field

TABLE S1. All four fixed-gap response comparisons, using the central-difference step 0.005Ω. The final row is the first-order, single-ε' continuum reference from Ref. [7], not an exact lattice constraint.

$\widehat{m}_1/\Omega$	L	R_{32}	R_{42}
0.75	32	2.2236	1.5000
0.75	36	2.1097	1.3014
0.90	32	2.1154	1.2236
0.90	36	2.0969	1.1744
Continuum, $\eta \to 0$		$\simeq 1.97$	$\simeq 1.17$

ε' is

$$\frac{m_a}{m_1} = r_a + \eta s_a + O(\eta^2), \qquad (s_2, s_3, s_4) \simeq (0.67, 1.33, 0.79). \tag{S14}$$

These coefficients follow from approximate matrix elements obtained with the truncated conformal space approach. Their common sign cannot explain the oppositely signed second- and third-ratio deviations at the working point through a single first-order ε' perturbation. This does not exclude the asymptotic flow of Eq. (S4).

For the lattice response, Δ is retuned as V_2 changes so that $\widehat{m}_1$ remains fixed. Differentiating that constraint gives

$$0 = \partial_{V_2}\widehat{m}_1 + \frac{d\Delta}{dV_2}\partial_\Delta \widehat{m}_1, \qquad \frac{d\Delta}{dV_2} = -\frac{\partial_{V_2}\widehat{m}_1}{\partial_\Delta \widehat{m}_1}. \tag{S15}$$

The measured ratio derivatives and their relative responses are

$$D_a = \left(\partial_{V_2} - \frac{\partial_{V_2}\widehat{m}_1}{\partial_\Delta \widehat{m}_1}\partial_\Delta\right)\widehat{r}_a, \qquad R_{32} = D_3/D_2, \qquad R_{42} = D_4/D_2. \tag{S16}$$

If a single small η controls the ratio changes in the scaling limit, $D_a \simeq s_a\, d\eta/dV_2$ along the constrained path. The unknown coupling derivative cancels, giving

$$R_{32} \simeq s_3/s_2 \simeq 1.97, \qquad R_{42} \simeq s_4/s_2 \simeq 1.17. \tag{S17}$$

The quoted ratios use the unrounded approximate matrix-element inputs, and inherit their accuracy; no subpercent theoretical precision is claimed. The denominator must be nonzero for these response ratios to be defined.

For $\widehat{m}_1/\Omega = 0.75, 0.90$, the selected centers are $V_2/\Omega = -1.500448, -1.446096$, respectively, chosen from the existing $\theta = 110^\circ$ scan before the response calculation. At each size, Δ is retuned at the center and at interaction offsets $\pm 0.005\Omega, \pm 0.010\Omega$, totaling 20 points across the two gaps and two sizes. The two central-difference step sizes agree within 0.012%. The smaller step agrees within 0.004% with Hellmann–Feynman derivatives [20]. For either Hamiltonian parameter λ,

$$\partial_\lambda \widehat{m}_a = \langle a|\partial_\lambda H|a\rangle - \langle 0|\partial_\lambda H|0\rangle. \tag{S18}$$

The ground-state derivative is therefore included in every gap derivative.

The maximum fixed-gap error is below $4 \times 10^{-11}\Omega$; all recorded eigenpair residuals are below $5 \times 10^{-12}\Omega$. Squared overlaps of continuously tracked candidate states exceed 0.9996. These checks support numerical continuity, not continuum particle identity. The maximum final multi-state residual is approximately $2.17 \times 10^{-12}\Omega$; the $5 \times 10^{-12}\Omega$ bound above also covers recorded intermediate root-solving states.

Between $L = 32$ and 36, the relative changes of (R_{32}, R_{42}) are $(0.89, 4.19)\%$ at gap 0.90Ω, and $(5.40, 15.26)\%$ at 0.75Ω, with $L = 36$ as denominator. Neither gap satisfies the preselected 3% size criterion for both ratios. Lowering the gap worsens agreement with the continuum reference and size drift. The close value of R_{42} at $L = 36$, 0.90Ω does not establish a universal response. Fixing the lowest gap fixes neither the velocity nor irrelevant couplings. These contributions are not separated at the available sizes.

S5. OPEN CHAINS, REFLECTION, AND LEAKAGE

For an open chain, define

$$O_{f,q} = \sum_{j=0}^{L-1} f_j e^{iqj} n_j, \qquad f_j = 1 \quad \text{or} \quad f_j = \sin^2\left[\frac{\pi(j+1)}{L+1}\right]. \tag{S19}$$

Here $q = 0, \pi$ labels a drive pattern, not a conserved momentum. Weights are divided by $\sum_j f_j^2$. With the tapered drive at $L = 27$,

$$(\omega_2^{\mathrm{o}}, \omega_3^{\mathrm{o}}, \omega_4^{\mathrm{o}})/\omega_1^{\mathrm{o}} = (1.2612, 1.9030, 1.9488), \qquad \omega_1^{\mathrm{o}} = 0.9438\Omega. \tag{S20}$$

These ratios are within 2% of E_7. A standing-wave estimate is

$$E_a(n) = \sqrt{\widehat{m}_a^2 + \left[\frac{nv\pi}{L+1}\right]^2}. \tag{S21}$$

Using periodic $L = 36$ gaps and the critical velocity, the $n = 1$ estimates at $L = 27$ differ from the first three lines by less than 0.8%, and from the fourth by 1.8%. This is a consistency check, not an independent identification of particles or boundary states.

For spatial reflection $R : n_j \mapsto n_{L-1-j}$ and a symmetric envelope, reindexing the sum gives

$$RO_{f,q}R^{-1} = e^{iq(L-1)} \sum_{j=0}^{L-1} f_j e^{-iqj} n_j. \tag{S22}$$

At the two drive patterns, this reduces to

$$RO_{f,0}R^{-1} = O_{f,0}, \qquad RO_{f,\pi}R^{-1} = (-1)^{L-1} O_{f,\pi}. \tag{S23}$$

For reflection eigenstates $|0\rangle, |n\rangle$ with parities p_0, p_n and a probe of parity p_O, a nonzero matrix element requires $p_n p_0 p_O = 1$. Thus even L gives opposite probe parities and forbids both probes from exciting the same reflection eigenstate from a common ground state. At odd L, both probes are reflection even, so shared lines are allowed.

Leakage means the unintended-to-intended weight ratio for a candidate *eigenstate*. At $L = 27$, tapering reduces A_1 leakage from 0.15% to 0.027%, but increases A_4 leakage from 1.4% to 1.94%. Reflection selection does not eliminate nearby states in a channel. At $L = 28$, the tapered A_2 response splits into lines at 1.250 and $1.266\,\omega_1^{\mathrm{o}}$ with weights 27 : 73. At $L = 27$, a staggered satellite near $1.160\,\omega_1^{\mathrm{o}}$ is compatible with an A_1 standing-wave excitation, but its boundary origin is not excluded. Tapering reduces its weight from 13% to 1.7% of the main line.

TABLE S2. Same-state leakage in percent for tapered open-chain probes at the reference point. The intended probe is staggered for A_1, A_3 and uniform for A_2, A_4. Even-L zeros follow from reflection symmetry; they are not numerical bounds on overlap of broadened frequency windows.

Candidate	$L = 19$	23	27	26	28
A_1	0.228	0.074	0.027	0	0
A_2	4.53	0.009	0.019	0	0
A_3	39.4	17.8	7.46	0	0
A_4	8.98	4.42	1.94	0	0

S6. THRESHOLDS, HEAVY STATES, AND EXPERIMENTAL SCALES

In the integrable, translation-invariant theory, the lowest even and odd two-particle thresholds are $2m_1$ and $m_1 + m_2$, from A_1A_1 and A_1A_2, respectively. Their matrix-element-weighted onsets are [12]

$$S_{\text{even}}(\omega) \propto (\omega - 2m_1)^{1/2}, \qquad S_{\text{odd}}(\omega) \propto (\omega - m_1 - m_2)^{-1/2}, \tag{S24}$$

just above the respective edges. The scattering amplitude $S_{11}(0) = -1$ forces the identical-particle form factor to vanish at threshold; the relevant mixed-particle form factor is nonzero. The S_{11} notation here denotes a scattering amplitude, distinct from the density spectrum S_k in Eq. (S6).

At the reference point, the next odd level moves from 2.340 to $2.284\widehat{m}_1$ between $L = 18$ and 36, toward $1 + \widehat{r}_2 = 2.268$. The next even level moves from 2.296 ($L = 24$) to $2.105\widehat{m}_1$, toward 2. These discrete trends do not verify Eq. (S24). The elevated first even scattering candidate also overstates the separation of A_4 from the continuum in a finite periodic chain.

Combined energy and weight trends do not reliably resolve A_5, A_6, A_7 across $L = 18$–36, $\theta = 110°$, and $t = 0.2, 0.3$. Their ideal masses lie above symmetry-allowed thresholds: A_5, A_7 above $2m_1$ and A_6 above $m_1 + m_2$. Form-factor perturbation theory can treat integrability-breaking decays [21]; allowed E_7 channels include $A_5 \to A_1A_1$ and $A_6 \to A_1A_2$ [7]. Finite closed-chain spectra do not measure decay widths, so the unresolved levels neither establish nor exclude larger-system heavy resonances.

The two-wavelength (515 and 813 nm) tweezer scheme in Ref. [3] is one proposed route to a sign-changing staggered drive, not yet demonstrated in that setup. For $\Omega/2\pi = 5.5$ MHz, the reference binding corresponds to $B_4/2\pi \simeq 0.25$ MHz. At the same lowest gap, the ideal E_7 binding would be about 0.15 MHz. Main-text Fig. 2 uses $m_{\text{ref}} = \widehat{m}_1(36) = 0.9196\Omega$, with each panel normalized to its own staggered peak height and no periodic size averaging. Its Gaussian standard deviations are $0.03m_{\text{ref}}$ (periodic) and $0.05m_{\text{ref}}$ (open). For the latter,

$$\text{FWHM} = 2\sqrt{2\ln 2}\,\sigma = 0.118m_{\text{ref}}, \qquad \text{FWHM}/2\pi \simeq 0.60 \text{ MHz}. \tag{S25}$$

These widths illustrate smoothing, not measured instrumental linewidths. The gray threshold markers use $2m_{\text{ref}}$ and $m_{\text{ref}} + \widehat{m}_2(36)$, not measured open-chain continuum edges. A linewidth below the binding scale is a useful target for direct resolution; no signal-to-noise or joint-fitting analysis is included.

Finally, the calibration test scales V_2 by 1 ± 0.017 at fixed Δ, motivated by Ref. [3]. For the errors e_a in Eq. (S8), the maximum changes at $L = 36$ are 0.13, 0.43, and 0.20 percentage points for $a = 2, 3, 4$. These are changes of the E_7-normalized errors, not $100\Delta\widehat{r}_a$. The lowest gap changes by up to 4.5%; b_4 ranges from 0.0461 to 0.0530, compared with 0.0492 at the center. This single-parameter check is not a complete experimental error budget. Finite-blockade corrections, state preparation, decoherence, and readout remain unmodeled.

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